

Investigations on Polyplex Stability During the Freezing Step of Lyophilization Using Controlled Ice Nucleation—The Importance of Residence Time in the Low-Viscosity Fluid State

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ABSTRACT: The aim of the study was to comprehensively investigate the influence of the freezing step during lyophilization on the stability of gene-delivery particles in order to better understand particle stabilization during freezing. Particle size of plasmid/linear polyethylenimine (LPEI) polyplexes at two DNA concentrations and at increasing sucrose–DNA ratios was investigated separately as a function of freezing procedure, ice-nucleation temperature, residence time of the particles in a partially frozen state, or incomplete freezing. Using a numerical model, the increase in sucrose concentration and system viscosity and corresponding bimolecular reaction rates were theoretically estimated. Freezing with a temperature-hold step after ice nucleation negatively influenced particle stability. Ice-nucleation temperature had an impact only at low DNA concentrations. Particle stability was highly reduced during the early part of freezing ($<-3^{\circ}\text{C}$), especially at low shelf-ramp rates. In this phase, bimolecular reaction rates increase greatly at still low system viscosity. Below a critical temperature ($\leq -18^{\circ}\text{C}$) and at high system viscosity, no further particle aggregation occurred. In conclusion, the initial sample viscosity rather than the unfrozen volume and the residence time of particles in the low-viscosity state are the predominant factors in particle stabilization, which likely apply to aggregation in any system. © 2012 Wiley Periodicals, Inc. and the American Pharmacists Association *J Pharm Sci* 102:929–946, 2013

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INTRODUCTION

Nucleic-acid-based therapeutics are a promising new class of biopharmaceuticals.¹ The most common non-viral carriers for the nucleic acids are lipoplexes and polyplexes.¹ These nanoparticulate carriers are based on cationic lipids or cationic polymers that form condensed complexes with the negatively charged nucleic acids.¹ One general limitation in the long-term stability of nanoparticles, specifically nonviral vectors, is their tendency to aggregate in aqueous solutions, which is known to correlate with a loss in transfection efficiency of nucleic acid-based

systems.² Consequently, lyophilization has proven to be an attractive way for the production of long-term stable, transfection efficient gene-delivery formulations. Although lyophilization is generally considered a very gentle drying process, it involves two potential stresses: freezing and drying.³ These stresses typically cause damage of macromolecules and nanoparticles, unless appropriate stabilizers are added to the formulation.⁴

In a previous study, we found that the aggregation of plasmid/linear polyethylenimine (LPEI) polyplexes occurs predominantly during the freezing and not during the drying step of the lyophilization process.⁵ Similar observations were also reported for lipid/DNA complexes.^{6–8} In general, selected excipients, especially disaccharides like sucrose, are very efficient stabilizers when present above a critical mass ratio of stabilizer to gene-delivery particle.^{7,9–11}

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During freezing, there are two main stress factors that could impact the stability of gene-delivery particles: ice formation and cryoconcentration.^{12–14} Ice formation can threaten particle stability via their exposure to ice–liquid interfaces or because of mechanical damage by growing ice crystals.¹⁴ Cryoconcentration leads to both solute- and particle-rich liquid phase and might facilitate particle aggregation because of an increase in ionic strength, pH shifts, or simply particle crowding.^{4,7,14}

Several mechanisms addressing how cryoprotectants may stabilize nonviral gene-delivery particles during freezing have been discussed.^{2,15} The “preferential exclusion hypothesis,” which was established to explain protein stabilization, states that solutes are preferentially excluded from a protein surface leading to the formation of a stabilizing solvent layer.^{2,15,16} However, it is doubtful whether the “preferential exclusion hypothesis” can be adapted to nonviral vectors.¹⁵ According to the “glass formation or vitrification hypothesis,” nonviral vectors are entrapped in the amorphous glassy matrix that is formed when the sample is cooled below the glass-transition temperature of the maximally freeze-concentrated system (T'_g).^{2,15} The high viscosity of the glassy matrix immobilizes gene-delivery particles and, thus, prevents particle aggregation.^{2,15} However, the “vitrification hypothesis” alone is not able to explain the requirement of a critical mass ratio of stabilizer to gene-delivery particle.¹⁵

In a study on the freeze–thaw stability of lipid/DNA complexes, Allison et al.⁷ demonstrated that glass formation is not obligatory as sugars like glucose are also able to preserve particle size at temperatures above T'_g . Consequently, the “particle isolation hypothesis” was proposed, which is based on the fact that the separation of individual particles in the unfrozen fraction prevents particle aggregation during freezing.⁷ At higher stabilizer–particle mass ratios, the volume of the unfrozen fraction increases, leading to greater dilution and fewer particle–particle contacts.⁷ Moreover, if the system has very high viscosity, diffusion is too slow to form new particle–particle contacts on the relevant time scale. In a further study, Armstrong and Anchordouy¹⁷ showed that particle aggregation can result from the formation of ice, as well as from prolonged incubation in the frozen state above, but not below a critical temperature well above T'_g at which particles are already sufficiently immobilized in the freeze concentrate.¹⁷

Until now, no detailed study on the influence of different freezing parameters, such as ice-nucleation temperature or freezing rate, on the stability of nonviral gene-delivery particles is available. These parameters impact the concentration and temperature history and therefore the residence time in a concentrated but relatively fluid system and thus, might

also highly impact particle stability. Studies of the impact of freezing variations on the stability of nonviral gene-delivery particles are limited to several reports of fast freezing in a dry ice/alcohol bath, in liquid nitrogen, or in a -70°C freezer resulting in better preservation of lipoplexes compared with slow freezing in a -20°C freezer or at $-2.5^\circ\text{C}/\text{min}$ to -40°C .^{6,18}

However, these fast freezing methods are difficult to control and are not practical on large scale. Instead, shelf-ramp freezing, where the filled vials are placed on the shelves of the freeze-drier and the shelf temperature is decreased linearly with time, is the most commonly used freezing method.¹³ During shelf-ramp freezing, ice nucleation occurs stochastically, resulting in vial-to-vial or batch-to-batch heterogeneity and scale-up issues.¹³ Therefore, different methods based on ultrasound, electro-freezing, vacuum-induced surface freezing, an ice fog, or depressurization have been developed to control nucleation at the desired nucleation temperature.¹³ Depressurization is a rather newly established method that has gained substantial interest.^{19,20} Using this method, Bursac et al.¹⁹ reported larger pores, lower specific surface area (SSA), and improved lactate dehydrogenase recovery for samples frozen by depressurization at $\sim -3^\circ\text{C}$ with an ~ 15 -min hold step after nucleation, as compared with samples produced by shelf-ramp freezing.

The objective of this study was to systematically investigate the influence of different freezing procedures on the stability of plasmid/LPEI polyplexes, in order to get an extended understanding of critical freezing parameters and the underlying stabilization mechanisms during freezing of nonviral gene-delivery particles. The polyplexes were formulated at different plasmid DNA concentrations and at increasing sucrose–polyplex mass ratios. In the first step, the influence of the “standard” depressurization method, which refers to the use of a 15-min hold period after nucleation, in comparison with conventional shelf-ramp freezing on polyplex particle size was investigated. Second, the influence of the ice-nucleation temperature and the residence time of the particles above T'_g during freezing on polyplex particle size were studied individually with the aid of the depressurization method to control nucleation. Furthermore, the increase in sucrose concentration and sample viscosity during freezing was theoretically calculated and correlated with bimolecular reactions rates. These theoretical results were then compared with the experimental data. The identification of the critical parameters in freezing and a better understanding of the stabilization mechanisms involved during freezing should help to design less stressful lyophilization processes and lyophilized products with improved quality.

Table 1. Overview on Polyplex Formulations at DNA Concentrations of 10 or 50 µg/mL and Various Sucrose–DNA Mass Ratios or Corresponding Sucrose Concentrations (% w/v)

DNA Concentration: 10 µg/mL		DNA Concentration: 50 µg/mL	
Sucrose–DNA Mass Ratio	Sucrose Concentration (% w/v)	Sucrose–DNA Mass Ratio	Sucrose Concentration (% w/v)
2000	2	1200	6
4000	4	1600	8
6000	6	2000	10
8000	8	2400	12
10,000	10	2800	14
12,000	12	3200	16
14,000	14	3600	18
		4000	20

MATERIALS AND METHODS

Materials

The plasmid (pCMVLuc) was produced by PlasmidFactory (Bielefeld, Germany). LPEI (22 kDa) was synthesized by acid-catalyzed hydrolyzes from poly(2-ethyl-2-oxazoline) (Sigma–Aldrich, Steinheim, Germany) as described in Ref. 21. L-Histidine was obtained from Merck (Darmstadt, Germany) and sucrose was obtained from Sigma–Aldrich (St. Louis, Missouri). Glass vials (3 mL, glass type: 8412-G) with their according LyoTec™ stoppers were used (West Pharmaceutical Services, Lionville, Pennsylvania).

Sample Preparation

Plasmid/LPEI polyplexes were prepared at a N:P ratio [molar ratio of LPEI nitrogen (N) to DNA phosphate (P)] of 6:1 and at a DNA concentration of 100 µg/mL in 10 mM L-histidine buffer pH 6.0 with an established micromixer method²² using a 5.0-mL syringe and a plunger speed of 1.0 cm/min. Polyplexes were incubated for 30 min at room temperature to ensure complete polyplex formation. Sucrose stock solutions at various concentrations (% w/v) were prepared in 10 mM L-histidine buffer pH 6.0. Polyplex formulations at DNA concentrations of 10 or 50 µg/mL and at various sucrose–DNA mass ratios (Table 1) were prepared by mixing the polyplex stock solution with the sucrose stock solutions.

Freeze–Thaw Studies Using Different Freezing Protocols

Freeze–thaw studies were performed on a pilot-scale freeze-drier (FTS LyoStar™ 3, SP Scientific, Stone Ridge, New York) equipped with the ControLyo™ technology (Praxair, Burr Ridge, Illinois) for ice-nucleation temperature control. Each polyplex formulation (500 µL) in 3-mL vials was investigated in triplicate in each freeze–thaw experiment. Vials were placed on the middle of the shelf in a hexagonal array, and all edge vials were surrounded by placebo vials containing a 5% sucrose solution. All samples were equilibrated at 20°C for 15 min prior to freezing. In all complete freeze–thaw experiments, samples

were frozen to a final shelf temperature of –45°C and held at this temperature overnight for 10 h. Thawing was always performed by a shelf-ramp rate of 20°C at 2.5°C/min with a 60-min hold at 20°C. To monitor product-temperature profiles during freezing, thin wire T-type thermocouples (Omega, Stamford, Connecticut) were dipped into thermal grease and fixed with tape on the outside of selected vials close to the vial bottom. Product-temperature data and shelf-inlet temperature data were collected every minute.

Conventional Shelf-Ramp Freezing

Conventional shelf-ramp freezing was performed at a shelf-ramp rate of –1 or –5°C/min to –45°C.

“Standard” Depressurization Method

The “standard” depressurization method for controlled ice nucleation was performed according to the method described by Konstantinidis et al.²⁰. Samples were cooled at a shelf-ramp rate of –1°C/min to 5°C and equilibrated at that temperature for 15 min. Air from the product chamber was purged once by pressurizing at 10 psig (171 kPa) for 2 min with argon gas and depressurizing to a final pressure of 2 psig (115 kPa) within 20 s. The product chamber was pressurized with argon gas to approximately 28 psig (294 kPa). Subsequently, samples were cooled at –1°C/min to –4°C and equilibrated at that temperature for 30 min. To induce ice nucleation, the chamber pressure was quickly released to 2 psig (115 kPa). After ice nucleation, the shelf temperature was held for another 15 min at –4°C. Subsequently, samples were cooled at a shelf-ramp rate of –1°C/min to –45°C. The aforementioned parameters applied during purging, pressurization, and depressurization of the product chamber were used in all further freeze–thaw experiments.

Varying Ice-Nucleation Temperatures Via Controlled Ice Nucleation

To vary ice-nucleation temperature during freezing, samples were cooled at –1°C/min to –4°C, –8°C, or –12°C after pressurization and equilibrated at that

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temperature for 30 min before ice nucleation was induced. Subsequently, samples were directly cooled at $-5^{\circ}\text{C}/\text{min}$ to -45°C without any further equilibration step.

Varying Shelf-Ramp Rates after Controlled Ice Nucleation

After pressurization, samples were cooled at $-1^{\circ}\text{C}/\text{min}$ to -12°C and equilibrated at that temperature for 30 min before ice nucleation was induced. Subsequently, samples were directly cooled at varying shelf-ramp rates of $-0.1^{\circ}\text{C}/\text{min}$, $-0.5^{\circ}\text{C}/\text{min}$, $-1.0^{\circ}\text{C}/\text{min}$, or $-5.0^{\circ}\text{C}/\text{min}$ to -45°C without a postnucleation temperature hold.

Interruption of the Freezing Process After Controlled Ice Nucleation

Samples were cooled at $-1^{\circ}\text{C}/\text{min}$ to -12°C after pressurization and equilibrated at that temperature for 30 min before ice nucleation was induced. Then, samples were directly cooled at $-0.1^{\circ}\text{C}/\text{min}$ to a shelf temperature of -12.5°C , -14.0°C , -20.0°C , and -26.0°C and immediately thawed at $2.5^{\circ}\text{C}/\text{min}$.

Lyophilization

Lyophilization was performed on the same pilot-scale freeze-drier as described in section *Freeze-Thaw Studies Using Different Freezing Protocols*. Selected polyplex formulations ($500\ \mu\text{L}$) in 3-mL vials were frozen to -45.0°C , using different freezing protocols, and held at -45.0°C for 60 min. For primary drying, the shelf temperature was increased at $0.2^{\circ}\text{C}/\text{min}$ to -30.0°C and the pressure was reduced to 65 mTorr (8.7 Pa). After $\sim 18\ \text{h}$, the endpoint of primary drying was confirmed by comparison of Pirani gauge and capacitance monometer readouts. Secondary drying was performed at 40°C (shelf-ramp rate of $0.1^{\circ}\text{C}/\text{min}$) and 65 mTorr (8.7 Pa) for 6 h. Samples were stoppered under vacuum.

Polyplex Characterization

Size determination of siRNA polyplexes was performed by dynamic light scattering using the Nano-sizer ZS (Malvern Instruments, Westborough, Massachusetts). The samples were diluted 1:20 with 10 mM L-histidine buffer pH 6.0 to a final volume of 1 mL. The diluted samples were analyzed in single-use polystyrene semimicro PMMA cuvettes (Brand, Wertheim, Germany) with a path length of 10 mm. After an equilibration time of 1 min at 20°C , each sample was recorded three times with five subruns of 10 s, using the multimodal mode. The Z-average diameter was calculated from the correlation function

using the Zetasizer Software 6.20 (Malvern Instruments, Westborough, Massachusetts).

SSA Determination

The SSA of selected lyophilizates was determined via BET surface area analysis using a Flowsorb II 2300 instrument (Micromeritics, Norcross, Georgia). Prior to each measurement, the instrument was calibrated via injection of 1 mL of 100% krypton gas at known ambient temperature and pressure. Approximately 100 mg of the lyophilizates was transferred into the sample holder and sealed with sealing film in a dry bag, purged with nitrogen, and then, connected to the instrument. For out gassing, samples were heated on a heating mantle at 40°C for 2 h in a gas flow of a helium and krypton mixture. No difference in SSA results was observed if samples were out gassed for more than 2 h. Subsequently, samples were exposed to a mixture of helium (inert carrier gas) and krypton (adsorbate) (0.1 mol % krypton in helium). The adsorption at 0.1 mol % krypton was determined in single-point measurements and analyzed with the BET equation. For each formulation and freezing condition, three samples from independent vials were analyzed at least twice. From these results, the average SSA (m^2/g) along with the standard deviation was calculated. To include the effect of variable ice surface area in samples of equal volumes but of variable sucrose levels, the absolute SSA ($\text{m}^2/100\ \text{mL}$)²³ was calculated by

$$\text{absolute SSA} \left[\frac{\text{m}^2}{100\ \text{mL}} \right] = \text{SSA} \left[\frac{\text{m}^2}{\text{g}} \right] \cdot \% \text{ solute} \left[\frac{\text{g}}{100\ \text{mL}} \right] \quad (1)$$

Karl-Fischer Titration

The residual moisture content of the lyophilizates was determined by coulometric Karl-Fischer titration using a Metrohm 756 KF Coulometer (Metrohm, Herisau, Switzerland). About $500\text{-}\mu\text{L}$ dried methanol (HydranalTM-Methanol, Sigma-Aldrich, Steinheim, Germany) was injected into the sealed vial through the stopper, and the lyophilized samples were dissolved by slightly shaking the vial. The water content was determined by injecting about $300\ \mu\text{L}$ of the methanol samples into the titration solution (HydranalTM-Coulomat AG, Riedel-de Haen, Seelze, Germany) and was calculated as weight percent water in the solid. HydranalTM Water Standard 0.10 (Sigma-Aldrich, Steinheim, Germany) was used as reference standard. For each selected formulation and freezing condition, three vials were analyzed.

Theoretical Modeling of the Change in Sucrose Concentration and Viscosity During Freeze Concentration and Prediction of Biomolecular Reaction Rates

The change in sucrose concentration and viscosity during freeze concentration and corresponding relative reaction rates were theoretically modeled as previously published.²⁴ In our study, the weight ratio of L-histidine buffer to sucrose was very small, so the effect of buffer was neglected. At first, the reduction of product temperature with each minute increment was simulated via estimation of the heat transfer from the shelf to the product and subsequent evaluation of the amount of ice formed.

The heat-transfer rate from the shelf to the product, Q_s (cal/s), is given by

$$Q_s = A_v K_{v(P)} (T_s - T_p) \quad (2)$$

where A_v and K_v are the cross-sectional area and heat-transfer coefficient at a defined pressure (P) of the vial, respectively, and T_s and T_p are shelf temperature and product temperature, respectively.²⁵ A_v was $\sim 2.19 \text{ cm}^2$. The vial heat-transfer coefficient K_v [cal/($\text{m}^2 \text{Ks}$)] is composed of three terms: the contact, K_c , the radiation, K_r , and the gas conduction term, K_g . The following constants for typical tubing vials were used: $10^4 K_c \sim 1.0$, $10^4 K_r \sim 1.5$, and $10^4 K_g \sim 4.3$ (at atmospheric pressure, with a mean separation distance of 1 mm).

The ratio R of the heat-transfer rate needed to remove heat created by the phase change (Q_{pc}) (i.e., forming of ΔX_w g of water) to the heat-transfer rate needed to cool both the solution and the ice formed (Q_{cool}) by the temperature change in the time interval Δt is calculated by

$$R = \frac{Q_{pc} \Delta t}{Q_{cool} \Delta t} \quad (3)$$

The heat-transfer rate needed to freeze ΔX_w g of water is given by

$$Q_{pc} \Delta t = \Delta X_w \Delta H_f \quad (4)$$

where ΔH_f is the heat fusion of ice (79.72 cal/g). The heat-transfer rate for cooling both the solution and the ice formed is given by

$$Q_{cool} \Delta t = (X_w C_{\text{water}} + (X_w^0 - X_w) C_{\text{ice}} + m_{\text{vial}} C_{\text{glass}}) \Delta T_{\Delta t} \quad (5)$$

where X_w^0 is the initial mass of water and X_w is the mass of water that is still present. C_{water} , C_{ice} , and C_{glass} are the specific heats of water, ice, and the glass

vials, which are 0.99, 0.49, and 0.20 cal/gK, respectively. The mass of the vials (m_{vial}) was ~ 5.15 g. $\Delta T_{\Delta t}$ is the change in temperature of the sample in the time interval Δt , and was calculated from the difference in the freezing point depression ΔT_f between each time interval. The freezing point depression ΔT_f is given by an empirical equation found to fit the data in²⁶:

$$\Delta T_f = 2.54 \times \frac{m_{\text{solute}}}{X_w} \times \exp(0.259 \times \frac{m_{\text{solute}}}{X_w}) \quad (6)$$

where m_{solute} is the mass of sucrose in g per vial. This calculated product temperature of the sample and not the product temperature measured with the thermocouples on the outside of the vial was used to model the change in sucrose concentration, viscosity, and relative reaction rates during freeze concentration.

Using the calculated factor R and Eq. 2, the heat provided from the shelf to form ice was determined. Thus, the amount of formed ice ΔX_w during each time interval could be calculated from Eq. 4, and the amount of unfrozen water X_w in the sample at each time point could be determined. For each sample composition, the temperature of the solution could be calculated from the liquidus curve, Eq. 6. From the value of X_w , the concentration of sucrose at each time point was calculated, until the maximum freeze concentration of sucrose of $\sim 80.9\%$ is reached.²⁷

In the next step, a relationship between the variables, product temperature and sucrose concentration and the viscosity of the sucrose sample was developed. For this purpose, viscosity data of sucrose solutions at various concentrations and temperatures given in the CRC Handbook²⁶ and quoted by Angell et al.²⁸ were used. To obtain an accurate viscosity estimation outside the available data range, the data were fitted with a Vogel–Tammann–Fulcher-type equation:

$$\ln(\eta) = \ln(A) + \frac{K}{T - T_0} \quad (7)$$

This equation relates viscosity (η_{poise}) to the absolute temperature (T) and the “zero mobility” temperature T_0 , where A is a constant, which should be 1×10^{-4} .²⁸ T_0 depends on the composition of the system as T_0 is related to the glass-transition temperature T_g . As an empirical equation to represent the composition dependence of T_0 , we chose a variation of the Fox equation often used to represent the composition dependence of T_g ,

$$\frac{1}{T_0} = \frac{1}{T_{00}} + b_1 c + b_2 c^2 \quad (8)$$

where T_{00} is the “zero mobility” temperature for pure water (i.e., when $c = 0$). According to Angell et al.,²⁸ $T_{00} = 125$ K. Moreover, b_1 and b_2 are arbitrary

constants to be determined by a fit to the data. For the constant K , which may also be dependent on the system composition, a simple linear relation to the concentration c was assumed as:

$$K = K_0 + k_1c \quad (9)$$

Thus, using the last three equations with the fit parameters $K_0 = 751$, $k_1 = 855$, $b_1 = 0.001551$, and $b_2 = -0.00617$, viscosity over the entire range of concentrations and temperatures during freeze concentration could be calculated.

To estimate the relationship between reaction rate and viscosity, a second-order reaction, with an Arrhenius temperature dependence and an activation energy of 20 kcal/mol, was assumed. This activation energy is arbitrary but representative. Moreover, reaction rate was coupled to viscosity, as a diffusion-controlled reaction was assumed, with the diffusion constant being inversely proportional to viscosity, as in the Stokes–Einstein equation. Polyplex concentration was calculated on the basis of the assumption that one polyplex contains five plasmid strands each consisting of 6233 base pairs. The relative reaction rate was related to the reaction rate at the point of ice nucleation at a DNA concentration of 50 $\mu\text{g/mL}$, which was defined as 1.

Statistical Analysis

Differences between mean particle sizes were statistically analyzed by performing a two-tailed student t test using the QuickCalcs, GraphPad Software (La Jolla, California). Mean values with a p value of less than 0.05 were judged as significantly different.

RESULTS

Comparison of the “Standard” Depressurization Method With the Conventional Shelf-Ramp Freezing on Polyplex Stability

As aforementioned, it is reported that superior stability of lactate dehydrogenase was observed for samples frozen with the “standard” depressurization technique compared with stochastic shelf-ramp freezing.¹⁹ Thus, as a first step, polyplex stability using the “standard” (i.e., 15-min postnucleation hold) controlled nucleation was compared with conventional shelf-ramped freezing. Therefore, polyplex samples formulated at 10 and 50 $\mu\text{g/mL}$ DNA and at increasing sucrose–DNA mass ratio (Table 1) were frozen via depressurization-induced ice nucleation at -4°C , with a 15-min equilibration step and at a subsequent shelf-ramp rate of -1°C/min , or via conventional shelf-ramp (uncontrolled) freezing at a shelf-ramp rate of -1°C/min or -5°C/min . After thawing,

plasmid/LPEI polyplex samples were analyzed for particle size.

The two most-divergent product-temperature profiles obtained during each of these three different freezing procedures are displayed in Figure 1a. Using conventional shelf-ramp freezing, ice nucleation occurs stochastically at varying product temperatures ($-12.4 \pm 1.0^\circ\text{C}$ or $-9.7 \pm 1.6^\circ\text{C}$) and varying points in time. In contrast, when ice nucleation is induced in a controlled manner, using the depressurization technique, ice nucleation occurs at almost the same product temperature ($-2.3 \pm 0.1^\circ\text{C}$) and at the same point in time. Moreover, the three freezing protocols result in different residence times in the partially frozen state. Here, the residence time was defined as time period between ice nucleation and the point at which a product temperature of -30°C is reached. The residence time was 7.8 ± 0.5 min when samples were frozen at a shelf-ramp rate of -5°C/min and increased to 14.8 ± 3.6 min at a slower shelf-ramp rate of -1°C/min . In contrast, a markedly prolonged residence time of 51.8 ± 0.8 min was observed when the samples were frozen via our “standard” depressurization method (i.e., ice nucleation at -4°C , 15-min isothermal hold after nucleation, and subsequent shelf-ramp rate of -1°C/min).

The Z-average diameter of the plasmid/LPEI polyplexes after freeze–thaw using the three different freezing procedures is displayed in Figure 2. In general, particle size becomes better preserved with increasing sucrose–DNA ratio. At a fixed sucrose–DNA ratio, the “standard” depressurization method led to a higher increase in particle size compared with conventional shelf-ramp freezing. For instance, the size of plasmid/LPEI polyplexes at a concentration of 50 $\mu\text{g/mL}$ and at a sucrose–DNA ratio of 2000 increased from ~ 113 to ~ 153 or ~ 146 nm when frozen at a shelf-ramp rate of -1°C/min or -5°C/min , respectively (Fig. 2a). However, when frozen with the “standard” depressurization method, the increase in particle size to ~ 205 nm was significantly ($p < 0.05$) more pronounced. Moreover, the freezing procedure also impacts the critical sucrose–DNA ratio, at which particle size could be fully maintained. At a DNA concentration of 50 $\mu\text{g/mL}$ ($p > 0.05$), particle size was maintained at a sucrose–DNA ratio of 3200 or 2800 when frozen at a shelf-ramp rate of -1°C/min or -5°C/min , respectively. In comparison, a sucrose–DNA weight ratio of 4000 was required to significantly ($p > 0.05$) preserve particle size when samples were frozen with the “standard” depressurization method. Analogous trends were observed for plasmid/LPEI polyplexes at a DNA concentration of 10 $\mu\text{g/mL}$ (Fig. 2b). However, at this DNA concentration, particle size could not be fully maintained with any formulation and freezing protocol. As described above, there are two important factors that can be used to discriminate between

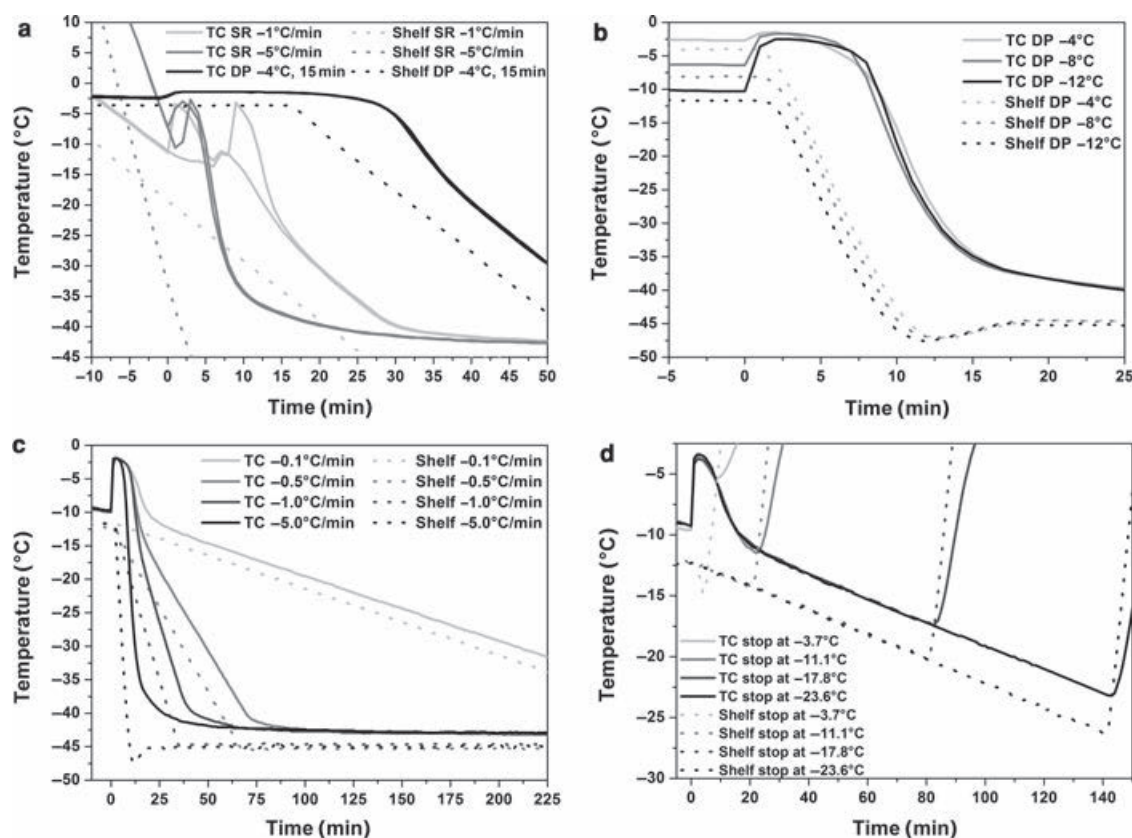


Figure 1. Product-temperature profiles of 10% sucrose samples monitored with thermocouples (TC) and inlet temperatures of the shelf (shelf) during freezing, using various freezing procedures. (a) Conventional shelf ramping (SR) at $-1^{\circ}\text{C}/\text{min}$ or $-5^{\circ}\text{C}/\text{min}$ or via the “standard” depressurization (DP) method at -4°C with a subsequent 15-min equilibration step and a shelf-ramp rate of $-1^{\circ}\text{C}/\text{min}$. The displayed product-temperature data represent the two most-divergent profiles obtained during the varying freezing procedures. (b) Nucleation induced by depressurization at a shelf temperature of -4°C , -8°C , or -12°C and a subsequent shelf-ramp rate of $-5^{\circ}\text{C}/\text{min}$. (c) Nucleation induced by depressurization at a shelf temperature of -12°C and subsequent varying shelf-ramp rates of $-0.1^{\circ}\text{C}/\text{min}$, $-0.5^{\circ}\text{C}/\text{min}$, $-1.0^{\circ}\text{C}/\text{min}$, or $-5.0^{\circ}\text{C}/\text{min}$. (d) Nucleation induced by depressurization at a shelf temperature of -12°C and subsequent shelf-ramp rate of $-0.1^{\circ}\text{C}/\text{min}$. The freezing procedure was stopped at a product temperature of -3.7°C , -11.1°C , -17.8°C , and -23.6°C .

freezing procedures: nucleation temperature and residence time. Thus, the influence of these two factors on polyplex particle size was investigated separately in the following experiments.

Influence of Ice-Nucleation Temperature on Polyplex Stability

To investigate the influence of ice-nucleation temperature on polyplex stability, the samples were equilibrated at various shelf temperatures (-4°C , -8°C , or -12°C) prior to depressurization. This resulted in product temperatures of $-2.2 \pm 0.2^{\circ}\text{C}$, $-6.3 \pm 0.3^{\circ}\text{C}$, or $-10.4 \pm 0.2^{\circ}\text{C}$ at the time of ice nucleation. Subsequently, samples were immediately cooled at $-5^{\circ}\text{C}/\text{min}$ to achieve residence times that do not significantly differ ($p > 0.05$) (13.2 ± 1.0 , 12.8 ± 0.9 , and 12.3 ± 0.6 min, respectively). The difference in ice-

nucleation temperature and the consistency in residence time are reflected by the monitored temperature profiles (as illustrated by the data for 10% sucrose samples in Fig. 1b).

At a DNA concentration of $50 \mu\text{g}/\text{mL}$ (Fig. 3a), no significant difference ($p > 0.05$) in particle size was observed at fixed sucrose–DNA ratios. For all ice-nucleation temperatures, complete retention of particle size was achieved at a sucrose–DNA ratio of 3200. At the lower DNA concentration of $10 \mu\text{g}/\text{mL}$ (Fig. 3b), particle size tended to be slightly better preserved when freezing was induced at lower ice-nucleation temperatures. However, this trend was only significant ($p < 0.05$) for all three ice-nucleation temperatures at sucrose–DNA ratios of 8000 and 10,000. Again, at the lower DNA concentration, complete particle stabilization could not be achieved.

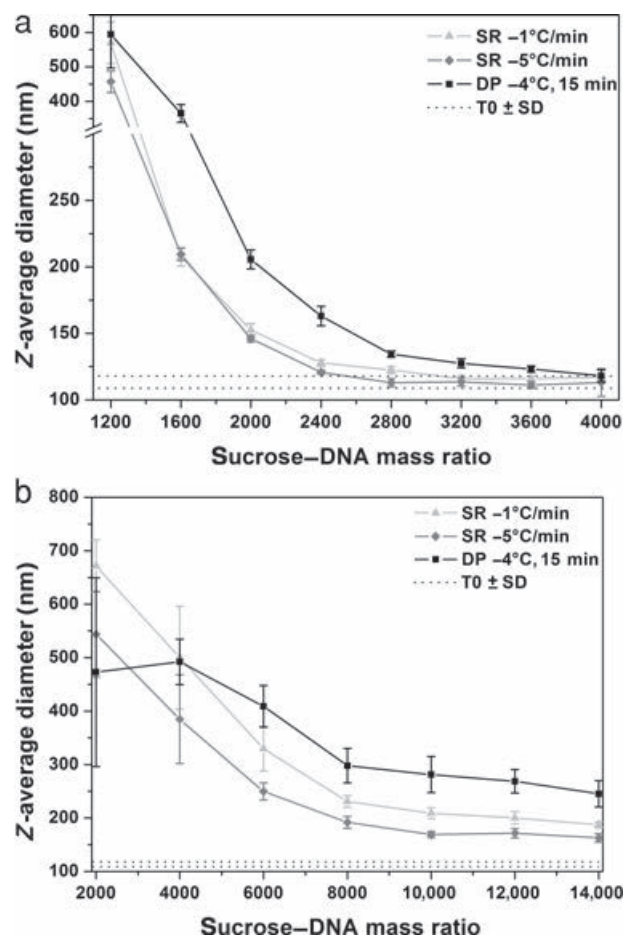


Figure 2. Z-average diameter of plasmid DNA/LPEI polyplexes after freeze-thaw when formulated at a DNA concentration of 50 µg/mL (a) or 10 µg/mL (b) and at increasing sucrose-DNA mass ratio. Samples were frozen via conventional shelf ramping (SR) at $-1^{\circ}\text{C}/\text{min}$ or $-5^{\circ}\text{C}/\text{min}$ or via the “standard” depressurization (DP) method at -4°C with a 15-min equilibration step and a subsequent shelf-ramp rate of $-1^{\circ}\text{C}/\text{min}$. The dotted lines represent the mean $\pm$ SD of the Z-average diameter of freshly prepared (T0) polyplexes. Each value represents the mean $\pm$ SD of three measurements on triplicate samples.

Influence of Residence Time on Polyplex Stability

In addition, the influence of residence time on polyplex stability was investigated. Here, ice nucleation of all polyplex samples was induced via depressurization after equilibration at a fixed shelf temperature of -12°C . Subsequently, samples were frozen at varying shelf-ramp rates of $-0.1^{\circ}\text{C}/\text{min}$, $-0.5^{\circ}\text{C}/\text{min}$, $-1.0^{\circ}\text{C}/\text{min}$, or $-5.0^{\circ}\text{C}/\text{min}$. The obtained temperature profiles are displayed in Figure 1c. The varying shelf-ramp rates after controlled ice nucleation resulted in greatly different residence times of 206.0 ± 3.4 , 48.5 ± 1.7 , 29.0 ± 0.8 , or 12.8 ± 0.5 min, respectively.

At a DNA concentration of 50 µg/mL (Fig. 4a), polyplex particle size did not significantly ($p > 0.05$) differ between samples frozen with shelf-ramp rates of

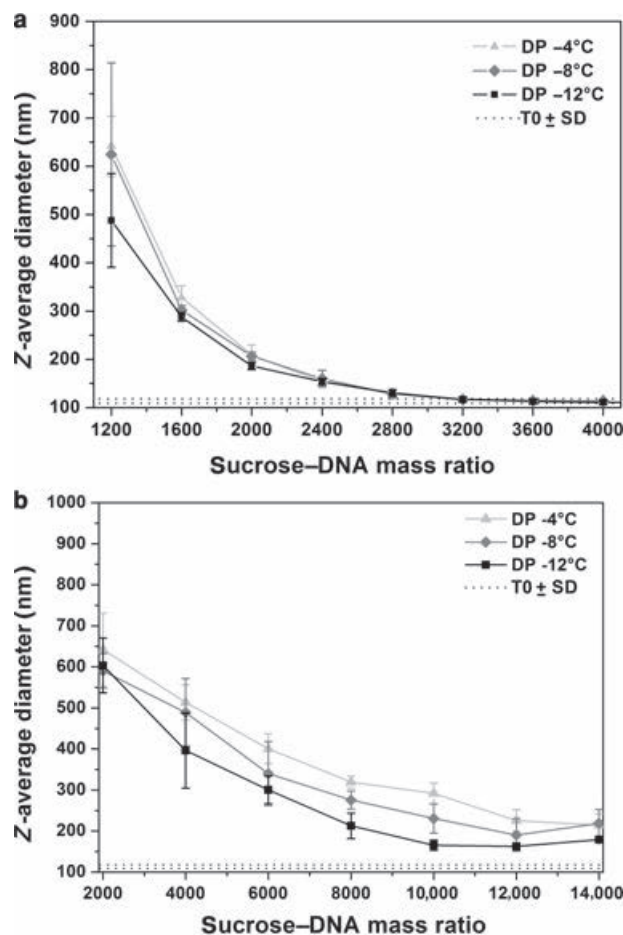


Figure 3. Z-average diameter of plasmid DNA/LPEI polyplexes after freeze-thaw when formulated at a DNA concentration of 50 µg/mL (a) or 10 µg/mL (b) and at increasing sucrose-DNA mass ratio. Samples were frozen at varying ice-nucleation temperatures induced by depressurization (DP) at a shelf temperature of -4°C , -8°C , or -12°C and a subsequent shelf-ramp rate of $-5^{\circ}\text{C}/\text{min}$. The dotted line represents the mean $\pm$ SD of the Z-average diameter of freshly prepared (T0) polyplexes. Each value represents the mean $\pm$ SD of three measurements on triplicate samples.

$-1^{\circ}\text{C}/\text{min}$ or $-5^{\circ}\text{C}/\text{min}$ after controlled ice nucleation. At fixed sucrose-DNA ratios, particle size increased with decreasing shelf-ramp rates from $-1^{\circ}\text{C}/\text{min}$ to $-0.5^{\circ}\text{C}/\text{min}$ or $-0.1^{\circ}\text{C}/\text{min}$. However, the variations between all of these three different freezing rates were only significant ($p < 0.05$) for the samples formulated at sucrose-DNA ratios between 2000 and 3200. Significant preservation ($p > 0.05$) of particle size was achieved at a sucrose-DNA ratio of 3200 when frozen at $-1^{\circ}\text{C}/\text{min}$ or $-5^{\circ}\text{C}/\text{min}$. At the lower shelf-ramp rate of $-0.5^{\circ}\text{C}/\text{min}$, a sucrose-DNA ratio of 4000 was required for the significant ($p < 0.05$) stabilization of particle size. However, when samples were frozen at $-0.1^{\circ}\text{C}/\text{min}$, particle size could not be completely stabilized within the investigated range of sucrose-DNA ratios. At a lower DNA concentration of 10 µg/mL,

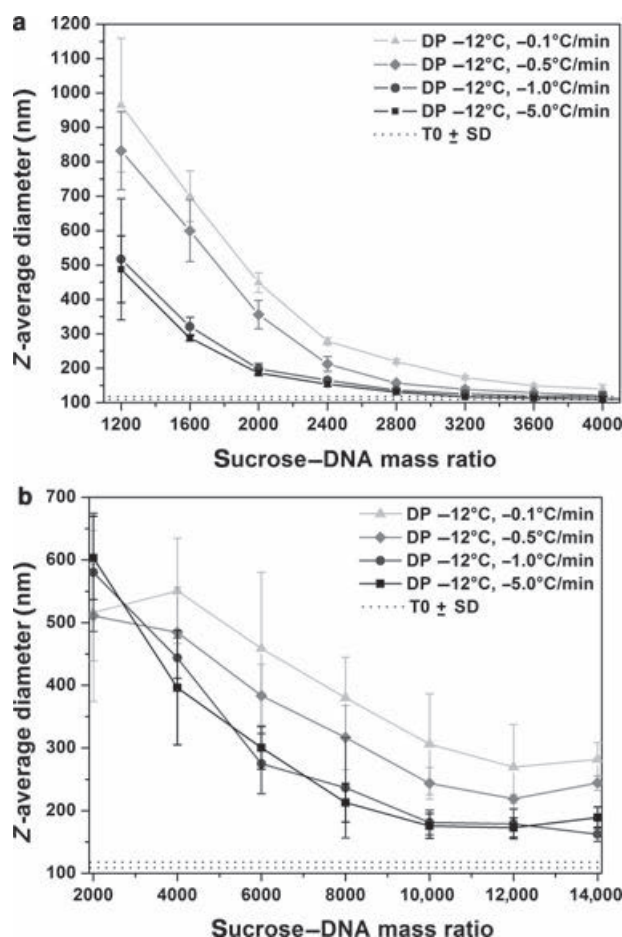


Figure 4. Z-average diameter of plasmid DNA/LPEI polyplexes after freeze-thaw when formulated at a DNA concentration of 50 µg/mL (a) or 10 µg/mL (b) and at increasing sucrose-DNA mass ratios. Samples were frozen with controlled ice nucleation via depressurization (DP) at -12°C and subsequent varying shelf-ramp rates of $-0.1^{\circ}\text{C}/\text{min}$, $-0.5^{\circ}\text{C}/\text{min}$, $-1.0^{\circ}\text{C}/\text{min}$, or $-5.0^{\circ}\text{C}/\text{min}$. The dotted line represents the mean $\pm$ SD of the Z-average diameter of freshly prepared (T0) polyplexes. Each value represents the mean $\pm$ SD of three measurements on triplicate samples.

particle size also did not significantly ($p > 0.05$) differ between samples prepared at shelf-ramp rates of $-1^{\circ}\text{C}/\text{min}$ or $-5^{\circ}\text{C}/\text{min}$. Moreover, an analogous trend with an increase in particle size when shelf-ramp rates are decreased was observed at the lower DNA concentration (Fig. 4b). This variation in particle size with differing shelf-ramp rates was only significant between samples that were frozen at $-1/-5^{\circ}\text{C}/\text{min}$ or $-0.1^{\circ}\text{C}/\text{min}$ when formulated at sucrose-DNA ratios ≥ 6000 . For instance, the decrease in the freezing rate from $-1^{\circ}\text{C}/\text{min}$ to $-0.1^{\circ}\text{C}/\text{min}$ resulted in a 120-nm increase in particle size after freeze-thaw (from 162 ± 12 to 282 ± 27 nm) when formulated at a sucrose-DNA ratio of 14,000. In general, for both DNA

concentrations, the increase in polyplex particle size was comparable when samples were frozen with the depressurization technique in combination with a fast shelf-ramp rate of either $-1^{\circ}\text{C}/\text{min}$ or $-5^{\circ}\text{C}/\text{min}$ or with conventional shelf-ramp freezing at shelf-ramp rates of either $-1^{\circ}\text{C}/\text{min}$ or $-5^{\circ}\text{C}/\text{min}$ (see the section *Comparison of the "Standard" Depressurization Method With the Conventional Shelf-Ramp Freezing on Polyplex Stability*).

Influence of Partial Freezing on Polyplex Stability

To investigate the influence of partial freezing, samples were immediately cooled at $-0.1^{\circ}\text{C}/\text{min}$ after controlled ice nucleation at -12°C , and the freezing process was stopped by subsequent thawing at shelf temperatures of -12.5°C , -14.0°C , -20.0°C , or -26.0°C . This approach resulted in partial freezing of the samples down to a product temperature of -3.7°C , -11.1°C , -17.8°C , or -23.6°C before thawing was started. The corresponding product temperature and shelf-inlet temperature profiles are displayed in Figure 1d.

At a DNA concentration of 50 µg/mL, particle size did not significantly ($p > 0.05$) differ, except for a sucrose-DNA ratio of 2400, when samples were frozen at a shelf-ramp rate of $-0.1^{\circ}\text{C}/\text{min}$ to a product temperature of -3.7°C or when samples were frozen at a shelf-ramp rate of $-5^{\circ}\text{C}/\text{min}$ to -45°C (as described in section *Influence of Residence Time on Polyplex Stability*) (Fig. 5a). In these two cases, complete preservation of particle size was achieved at a sucrose-DNA ratio of 3200. When the samples were further frozen at a shelf-ramp rate of $-0.1^{\circ}\text{C}/\text{min}$ to -11.1°C , particle size significantly increased further ($p < 0.05$) for all formulations at a sucrose-DNA ratios ≥ 1600 , and full preservation of particle size was observed at a sucrose-DNA ratio of 4000. Continuing freezing from -11.1°C to a product temperature of -17.8°C resulted in a further significant increase ($p < 0.05$) in particle size. However, when freezing proceeded further, from -17.8°C to product temperatures of $\sim -23.6^{\circ}\text{C}$, no significant ($p > 0.05$) further change in particle size was observed. For the latter two cases, freezing to -17.8°C and -23.6°C , complete polyplex stabilization could not be obtained within the investigated range of sucrose-DNA ratios. At the lower DNA concentration of 10 µg/mL, this correlation could not be clearly observed (Fig. 5b). Again, freezing at a shelf-ramp rate of $-0.1^{\circ}\text{C}/\text{min}$ to a product temperature of -3.7°C or freezing at a shelf-ramp rate of $-5^{\circ}\text{C}/\text{min}$ to -45°C resulted in comparable particles sizes after freeze-thaw. For all other investigated freezing procedures, an increase in particle size was produced, but no significant correlation with the extent of freezing was found.

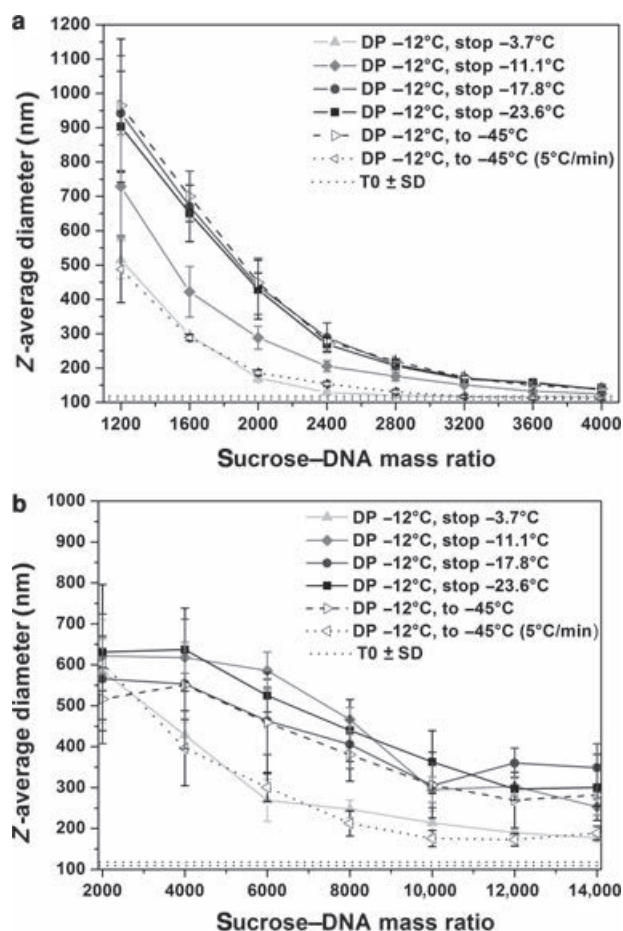


Figure 5. Z-average diameter of plasmid DNA/LPEI polyplexes after freeze-thaw when formulated at a DNA concentration of 50 µg/mL (a) or 10 µg/mL (b) and at increasing sucrose-DNA mass ratio. Samples were frozen with controlled ice nucleation via depressurization (DP) at -12°C and with a subsequent shelf-ramp rate of $-0.1^{\circ}\text{C}/\text{min}$. The freezing procedure was stopped at a product temperature of -3.7°C , -11.1°C , -17.8°C , and -23.6°C . For comparison reasons, data obtained for samples frozen with controlled ice nucleation via depressurization (DP) at -12°C and subsequent shelf-ramp rates of $-0.1^{\circ}\text{C}/\text{min}$ and $-5.0^{\circ}\text{C}/\text{min}$ are also displayed. The dotted line represents the mean $\pm$ SD of the Z-average diameter of freshly prepared (T_0) polyplexes. Each value represents the mean $\pm$ SD of three measurements on triplicate samples.

SSA and Residual Moisture Content of Selected Lyophilized Samples

Selected samples (6%, 10%, 14%, and 20% sucrose) were lyophilized using different freezing procedures to determine the impact of the freezing procedure on the resulting SSA or residual moisture content of the lyophilizates (Table 2). For all formulations, the SSA (m^2/g) decreased, the absolute SSA ($\text{m}^2/100\text{ mL}$) increased, and the residual moisture content increased with increasing sucrose concentration. In general, the freezing procedure highly influenced SSA and resid-

ual moisture. The increase in ice-nucleation temperature by 8°C (from -12°C to -4°C) at consistent subsequent shelf-ramp rates of $-5^{\circ}\text{C}/\text{min}$ resulted in a significant ($p < 0.05$) decrease in both SSA (m^2/g) and absolute SSA ($\text{m}^2/100\text{ mL}$) and a slight but not significant increase ($p > 0.05$, except for the 20% sucrose sample) in residual moisture for all formulations. The decrease in the shelf-ramp rate from $-5^{\circ}\text{C}/\text{min}$ to $-0.1^{\circ}\text{C}/\text{min}$ after controlled nucleation at -12°C led to significantly ($p < 0.05$) lowered SSA (m^2/g) and absolute SSA ($\text{m}^2/100\text{ mL}$) as well as a significantly ($p < 0.05$) higher residual moisture content for all formulations. Freezing of the samples using the "standard" depressurization method rendered the lowest SSA (m^2/g) and absolute SSA ($\text{m}^2/100\text{ mL}$) and the highest residual moisture content for all formulations.

Theoretical Modeling of the Change in Sucrose Concentration or Viscosity During Freeze Concentration and Correlation to Reaction Rates

To correlate the experimental observations with the increase in viscosity of the samples during freeze concentration, theoretical product-temperature profiles were modeled and used to calculate the change in sucrose concentration and sample viscosity over time. In addition, relative reaction rates were estimated assuming a second-order reaction with an Arrhenius temperature dependence, and relative reaction rates over time were estimated. The simulated temperature profiles and the calculated sucrose concentrations and viscosities of a 10% sucrose sample, that was nucleated at -12°C and subsequently frozen at a shelf-ramp rate of $-5^{\circ}\text{C}/\text{min}$ or $-0.1^{\circ}\text{C}/\text{min}$, are displayed in Figure 6. In general, the theoretically modeled temperature profiles showed rather good agreement with real product-temperature data (see Fig. 1). For instance, when freezing a 10% sucrose sample at a shelf-ramp rate of $-0.1^{\circ}\text{C}/\text{min}$, experimental product temperature was $-12.7 \pm 0.3^{\circ}\text{C}$ 30 min after ice nucleation or $-21.7 \pm 0.3^{\circ}\text{C}$ 120 min after ice nucleation in comparison with the calculated product temperature of -12.9°C or -24.5°C , respectively. When the initial ice nucleation occurs, product temperature rises rapidly to the equilibrium freezing point. Further ice-crystal growth is then controlled by the latent-heat release due to ice formation and the cooling rate of the shelf to which the sample is exposed. At first, heat release and cooling are in balance, leading to a temperature plateau. Later, when cooling of the shelf dominates, product temperature drops in a manner depending on the shelf-ramp rate used. During the first part of the freezing step, when ice formation and heat release are still in balance, most of the ice is formed and the temperature remains near the equilibrium freezing point of about -1°C . In this temperature region, the sucrose concentration of the sample sharply

Table 2. SSA (m²/g) or Absolute SSA (m²/100 mL) and RM Contents (% w/w) of Selected Lyophilized Formulations (6%, 10%, 14%, and 20% Sucrose)*

		6% Sucrose	10% Sucrose	14% Sucrose	20% Sucrose
DP -12°C, -5°C/min	SSA (m ² /g)	0.82 ± 0.07	0.78 ± 0.04	0.58 ± 0.02	0.51 ± 0.03
	SSA (m ² /100 mL)	4.92 ± 0.42	7.80 ± 0.40	8.12 ± 0.28	10.20 ± 0.60
	RM (% w/w)	0.19 ± 0.03	0.23 ± 0.04	0.35 ± 0.04	0.40 ± 0.04
DP -4°C, -5°C/min	SSA (m ² /g)	0.62 ± 0.03	0.44 ± 0.02	0.32 ± 0.01	0.30 ± 0.01
	SSA (m ² /100 mL)	3.72 ± 0.18	4.40 ± 0.20	4.48 ± 0.28	6.00 ± 0.20
	RM (% w/w)	0.20 ± 0.04	0.28 ± 0.02	0.42 ± 0.05	0.55 ± 0.02
DP -12°C, -0.1°C/min	SSA (m ² /g)	0.73 ± 0.04	0.59 ± 0.04	0.43 ± 0.03	0.40 ± 0.02
	SSA (m ² /100 mL)	4.38 ± 0.16	5.90 ± 0.40	6.02 ± 0.42	8.00 ± 0.40
	RM (% w/w)	0.38 ± 0.04	0.43 ± 0.09	0.44 ± 0.06	0.52 ± 0.05
DP -4°C, 15 min, -1°C/min	SSA (m ² /g)	0.51 ± 0.02	0.30 ± 0.01	0.23 ± 0.01	0.19 ± 0.01
	SSA (m ² /100 mL)	3.06 ± 0.12	3.00 ± 0.10	3.22 ± 0.14	3.80 ± 0.20
	RM (% w/w)	0.46 ± 0.03	0.49 ± 0.06	0.75 ± 0.02	0.83 ± 0.10

*The samples were frozen via DP at -12°C or -4°C at a subsequent ramp rate of -5°C/min or -0.1°C/min or via DP at -4°C with a 15-min equilibration step and a ramp rate of -1°C/min.

DP, depressurization; RM, residual moisture; SSA, specific surface area.

increases followed by a further slower increase until the maximum freeze concentration is reached at T_g' . For instance, a more than five-fold increase in the sucrose concentration from 10% to ~55% is already reached when the product temperature falls below -4°C after ice nucleation. When freezing proceeds to product temperatures of ~-12°C, sucrose concentration further increases to ~72%. At the product temperature of ~-18°C, at which no further increase in particle size is observed during slow freezing, the sucrose concentration is ~76%. Analogously, viscosity also tremendously increases during the first part of freezing. Because of the temperature dependence of the viscosity, at least as described by the model, a further pronounced increase is seen also at temperatures well below T_g' . For instance, viscosity values

increase to ~0.14, 20.2, 741.0, 53.3 × 10⁶, or 1.5 × 10¹² Pa·s when a product temperature of ~-4°C, -12°C, -18°C, -32°C, or -45°C is reached, respectively. Obviously, a faster increase in sucrose concentration and sample viscosity is observed when samples are frozen at a shelf-ramp rate of -5°C/min compared with freezing at a shelf-ramp rate of -0.1°C/min.

Figure 7 displays the estimated relative reaction rate over time and the corresponding viscosity curves at different shelf-ramp rates or different initial sucrose concentrations. During freezing of a 10% sucrose formulation (Fig. 7a), the reaction rate already increases five fold almost immediately after ice nucleation and increases more than 10 fold during further freezing. The maximum relative reaction rate is already reached when the sample viscosity is

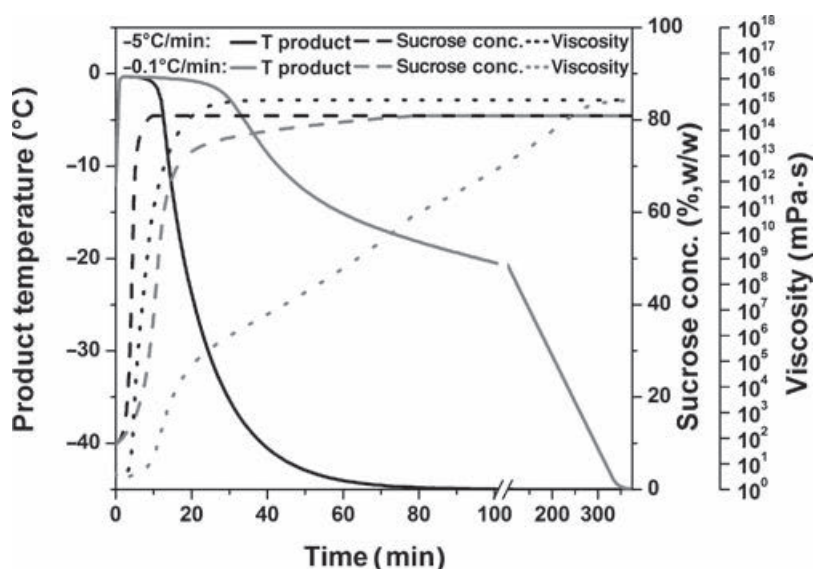


Figure 6. Theoretical simulation of product temperature (T_p), sucrose concentration (suc) and viscosity (η) of 500 μ L 10% sucrose solution in a 3-mL vial during shelf-ramp freezing at a shelf-ramp rate of -5°C/min or -0.1°C/min. Ice nucleation took place at time zero.

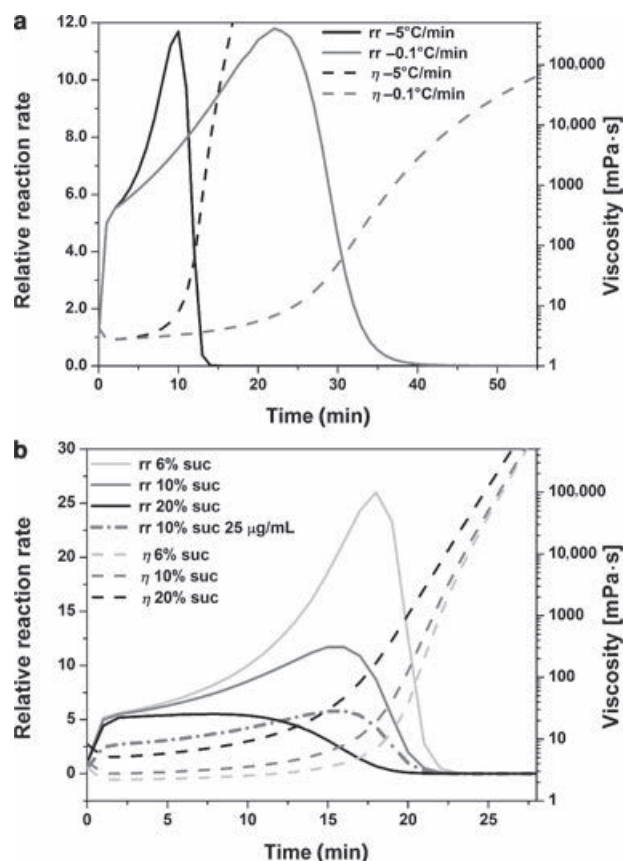


Figure 7. Theoretical relative bimolecular reaction rate (*rr*) of a second-order reaction of polyplexes during shelf-ramp freezing at a sample volume of 500 μ L in a 3-mL vial. Corresponding simulated viscosity values (η) are also displayed. Reaction rates were calculated with Arrhenius temperature dependence and inversely proportional correlated to modeled viscosity values. (a) Freezing of a 10% sucrose formulation at a DNA concentration of 50 μ g/mL at a shelf-ramp rate of $-5^{\circ}\text{C}/\text{min}$ or $-0.1^{\circ}\text{C}/\text{min}$. (b) Freezing of a 6%, 10%, or 20% sucrose formulation at a DNA concentration of 50 μ g/mL or of a “half concentrated” 10% sucrose formulation at a DNA concentration of 25 μ g/mL at a shelf-ramp rate of $-1^{\circ}\text{C}/\text{min}$. Ice nucleation took place at time zero.

~ 7 mPa·s at a product temperature of $\sim -1^{\circ}\text{C}$, which is essentially the equilibrium freezing point. When a sample viscosity of ~ 50 mPa·s is reached at a product temperature of $\sim -3^{\circ}\text{C}$, the relative reaction rate falls below the initial reaction rate in the liquid formulation at the point of ice nucleation. Subsequently, the relative reaction rates converge toward zero (<0.01) at a viscosity of $\sim 16,500$ mPa·s at a product temperature of $\sim -12^{\circ}\text{C}$. When samples are frozen at a shelf-ramp rate of $-5^{\circ}\text{C}/\text{min}$, the reaction rate decreases to the initial reaction rate after only ~ 12 min, whereas it takes ~ 30 min to reach this point when samples are frozen at $-0.1^{\circ}\text{C}/\text{min}$.

When 50 μ g/mL DNA samples with different initial sucrose concentration are frozen at a ramp rate of $-1^{\circ}\text{C}/\text{min}$ (Fig. 7b), the reaction rate also immedi-

ately increases $\sim$ five fold after initial ice nucleation for all formulations. During further freezing, the reaction rate increased drastically in presence of 6% and 10% to a maximum of relative reaction rate of 26.0 and 11.7, respectively. In contrast, the reaction rate did not increase further when samples contained 20% sucrose. This might be related to the already ~ 2.4 -fold higher viscosity of 20% sucrose formulations compared with the 6% samples at the point of ice nucleation. When polyplexes were formulated at higher sucrose concentrations, relative reaction rates returned to unity much earlier during freezing. However, for all formulations, reaction rates close to zero (<0.01) were found at almost the same time point ~ 24 min after ice nucleation. Additionally, the reaction rate for a polyplex formulation at half of the original DNA concentration containing 10% sucrose was modeled. In this case, the initial reaction rate was lower because of the reduced polyplex concentration. The maximum in reaction rate was reached at the same point in time as observed for the original 10% sucrose formulation. However, the level of the maximum reaction rate was comparable with the 20% sucrose formulation, which has the same initial sucrose–DNA ratio.

DISCUSSION

Several studies have demonstrated that stable transfection rates of nonviral gene-delivery particles are closely correlated with the maintenance of particle size.^{6,29} Even in the absence of reduced in-vitro transfection rates, particle size is a critical quality attribute for in-vivo application and overall for a pharmaceutical product.³⁰ Therefore, this study focused solely on particle size as an indicator of aggregation reflecting physicochemical instability, which can be affected by freeze–thaw and lyophilization. In general, aggregation of gene-delivery particles originates from colloidal mechanisms, involving electrostatic and/or van der-Waals interactions.^{2,31}

In accordance with previous studies,^{5–7,10} it was observed that the particle size of plasmid/LPEI polyplexes is drastically affected by freeze–thaw and that particle size became better preserved as the ratio of sucrose to plasmid DNA was increased. It is well known from the literature that maintenance of particle size requires a critical stabilizer–DNA ratio, which is highly dependent on the type of nonviral gene-delivery particle. For instance, the maintenance of particle size of LPEI,⁵ polyethylenimine,⁸ transferrin-polyethylenimine,⁹ or DMAEMA²⁹ polyplexes or DOTAP-DOPE,⁸ DOTAP-Chol,⁸ or DMRIE-C⁷ lipoplexes could be achieved at sucrose–DNA mass ratios of 4000, 7500, 10,000, or 1250, or 1250, 900, or 500, respectively. This existence of a critical stabilizer–DNA ratio led to the development of the “particle isolation hypothesis”⁷ stating that the

volume of the unfrozen fraction, in which particles are randomly suspended, must be only high enough to isolate the complexes in order to decrease the probability of particle collision. There is no doubt that particle crowding promotes aggregation during freezing and that particles have to be separated in order to inhibit particle aggregation. However, there are two observations in our study that indicate that the “particle isolation hypothesis,” based on the availability of a large-enough volume of the unfrozen fraction, is not suitable to solely explain the stabilization of nonviral gene-delivery particles during freezing.

First, it was observed that the stability of nonviral vectors upon freeze–thaw is not only dependent on the type of vector but also on the initial DNA or particle concentration of the sample. Moreover, it was found that the increase in particle size of plasmid/LPEI polyplexes was more pronounced in more dilute samples (10 μg DNA/mL) compared with more concentrated samples (50 μg DNA/mL) at the same sucrose–DNA ratio. In contrast to the more concentrated samples, for which complete particle stabilization could be achieved at a sucrose–DNA ratio of more than 2800, the complete preservation of particle size of more diluted plasmid/LPEI polyplexes was not even possible at sucrose–DNA ratios of 14,000. Similarly, Allison et al.⁷ reported that the freezing of more dilute samples of lipoplexes (8- μg DNA/mL) resulted in significantly larger particle sizes than freezing of more concentrated samples (160- μg DNA/mL) when formulated at the same sucrose–DNA ratio.⁷ However, in this case, the doubling of the sucrose–DNA ratio to 1000 was sufficient to stabilize the more diluted lipoplex samples.⁷

As in our study, the size of an individual gene-delivery particle was always consistent in the initial solution, independent of the DNA concentration; a lower DNA concentration can be equated with a lower particle concentration. Thus, the stabilizer–DNA ratio required for particle stabilization is expected to stay constant in accordance to the “particle isolation hypothesis,” independent of the initial particle concentration. However, this was not the case in our study, nor for plasmid/DMRIE-C lipoplexes, as reported by Allison et al.⁷ Other hypotheses that might explain the observed lowered particle stability in more diluted samples and the inability to completely stabilize low-concentrated plasmid/LPEI polyplexes will be discussed below.

Second, it was observed that the freezing procedure highly influences the freeze–thaw stability of plasmid/LPEI polyplexes. For instance, we found that freezing of plasmid/LPEI polyplexes by conventional shelf-ramp freezing resulted in a less pronounced increase in particle size compared with freezing by the “standard” depressurization method, which includes a 15-min isothermal hold after ice nucleation. How-

ever, the volume of the unfrozen fraction is independent of the freezing procedure and only controlled by the initial excipient concentration prior to freezing and the concentration of the maximum freeze concentrate, which is reported to be $\sim 81\%$ for sucrose.²⁷ Thus, particle stabilization during freezing cannot be explained solely by the available volume of the unfrozen fraction, but other parameters during freezing must be involved in stabilization.

When comparing the conventional shelf-ramp freezing with the “standard” depressurization method, two freezing parameters were identified that could explain the observed differences in particle stability: (1) ice-nucleation temperature and (2) residence time of the particles in the freeze concentrate until glass formation occurs. These two parameters were investigated separately by the aid of the “depressurization method,” which enables control of the ice-nucleation temperature. Moreover, various shelf-ramp rates after controlled ice nucleation were used to achieve different residence times. In general, the depressurization method is a very elegant, easy to apply method that can be used to guarantee homogeneous ice formation at a defined ice-nucleation temperature, in comparison with conventional shelf-ramp freezing during which ice nucleation occurs stochastically. In addition, in this study, the depressurization method was highly useful to discriminate between the effect of ice-nucleation temperature and subsequent freezing rates on the stability of an active pharmaceutical ingredient. Previously, only a few somewhat cumbersome methods were available to study the effect of ice-nucleation temperature on protein activity, such as seeding with ice crystals,²³ ultrasound-controlled ice nucleation,³² or the ice fog technique,³³ and systematic “mechanistic” studies are largely absent from the literature.

When different ice-nucleation temperatures at constant residence times were compared, the trend of a slightly better maintenance of particle size when freezing was induced at lower ice-nucleation temperatures was only observed for low-concentrated plasmid/LPEI polyplex formulations. In general, as the ice-nucleation temperature decreases, the number of formed ice crystals increases, thereby resulting in the formation of an increased ice–water interfacial area.¹³ For example, in our study, the decrease of ice-nucleation temperature from -4°C to -12°C results in an increase of the absolute ice surface per vial between 32% and 81%, depending on the initial sucrose concentration. This is in accordance with literature, in which an increase in the ice surface area by nearly 50% is reported when the ice-nucleation temperature was decreased from -1°C to -11°C .³⁴ Moreover, the observed increase of the SSA per sample vial with increasing solid content also agrees well with literature.²³ For proteins, it is well known that

surface-induced denaturation can be damaging³⁵ and that the difference in ice-nucleation temperature can greatly influence protein stability.^{23,36} Here, as one possibility, it is speculated that the mechanism involves very strong electric fields that are generated during ice crystallization due to preferential incorporation of one ionic species into the ice.^{24,37} The strength of the electric field depends on the freezing rate, the type of solvent, and the type and concentration of ionic species present in solution.^{24,38,39} Because in our study enhanced particle preservation was observed at lower ice-nucleation temperatures, which induces the formation of large ice–water interfaces, it is clear that interaction of the particles with the ice surface is not the only significant factor in particle destabilization. Rather, it is speculated that the aforementioned process-dependent generation of electric fields on the ice surface might contribute to the destabilization of the plasmid/LPEI particles, as these particles are predominantly held together by electrostatic interactions between the negatively charged nucleic acid and the positively charged polymer. However, why should the generation of electrically charged ice surfaces only significantly influence nonviral gene-delivery particles in diluted formulations when formulated at the same absolute amount of stabilizer (e.g., 14% sucrose)? Speculatively, it might be assumed that only a certain number of gene-delivery particles can interact with the ice–water surface during freezing and that this interaction plays a relatively larger role in more dilute samples. This is essentially saying that the ice–aqueous surface can saturate with particles at relatively low particles concentration, a common concept when interpreting interaction of proteins with surfaces. Consequently, one can tentatively conclude that for dilute samples, ice formation is one destabilizing factor for gene-delivery particles during freezing. In this case, it is not unexpected that non-surface-active excipients, like sucrose, even at high concentrations are not capable of fully stabilizing the particles. Analogously, Allison et al.⁷ stated that the reduced stability of lipoplexes in more dilute samples could be explained in general by an increased interaction with the ice-crystal surface area. This assumed interaction of the particles with the ice surface is additionally supported by Yu and Anchordoquy,⁴⁰ who reported that only the synergistic effects of surfactants and sugars are capable to stabilize dilute polyplex formulations. In contrast to our study, Allison et al.⁷ could achieve complete preservation of particle size for more dilute lipoplex samples. Presumably, as polyplexes are predominantly based on electrostatic interactions, whereas lipoplexes are additionally stabilized by hydrophobic interactions, polyplex stability might be more disturbed during the formation of charged ice interfaces as compared with the integrity of lipoplexes. Thus, this hypothesis also

explains the higher sugar levels needed for the stabilization of polyplexes compared with lipoplexes found in a head-to-head comparison study by Molina et al.⁸. Furthermore, our results give experimental evidence that in the case of more concentrated formulations, factors other than ice formation are responsible for the destabilization of gene-delivery particles during freezing.

To investigate the time dependence of particle stability during freezing, plasmid/LPEI polyplexes were frozen at different shelf-ramp rates after ice nucleation at a constant temperature. In general, for high- and low-concentrated polyplex formulations, particle size of the polyplexes and the critical sugar–DNA ratio that is required for full particle-size protection strongly increased with increasing residence times. Thus, not only the volume of the unfrozen fraction but also the time span during which a particle is exposed to freezing-induced stresses is of high importance. Moreover, these results give experimental evidence that particle stability is not only time dependent during incubation in the frozen state as reported by Armstrong and Anchordoquy¹⁷ but is also highly time dependent already during the freezing process. This negative effect of prolonged residence times in the critical-temperature region during freezing also explains why the “standard depressurization technique,” which involves a temperature-hold step after ice nucleation as described in Ref. 20, led to a more pronounced increase in particle size compared with conventional shelf-ramp freezing at fast freezing rates. However, the combination of the depressurization technique with subsequent fast freezing rates resulted in comparable particles size after freezing. As fast freezing rates reduce the particles’ residence time in the critical temperature range, quench freezing in liquid nitrogen or dry ice/ethanol might be advantageous for the retention of particle size during freezing. However, such uncontrolled freezing methods promote heterogeneous ice-crystal formation, potentially lead to a decrease in freeze drying rates, and are in general not applicable in large-scale manufacturing.¹³

It was found that the decrease in the shelf-ramp rate after controlled nucleation resulted in a decrease in the SSA, although ice-nucleation temperature was kept constant. Theoretically, the number of ice crystals that is initially formed during nucleation is defined by the ice-nucleation temperature and thus, determines the resulting ice surface area. But our results show that besides ice-nucleation temperature, the freezing rate also influences the resulting surface area. It can be speculated that the slow freezing rate leads to a situation that is comparable with a short annealing phase, during which larger crystals grow at the expense of smaller ones,⁴¹ leading to a reduced SSA of the product.⁴²

At first, we defined the residence time as the period of time between ice nucleation and the point at which a product temperature of -30°C is reached. The latter temperature is around the T_g' of the glassy sucrose matrix,⁴³ below which particle mobility should be highly limited. However, further experiments showed that freezing at very slow shelf-ramp rates is only critical with regard to particle stability until a product temperature of $\sim -18^{\circ}\text{C}$ is reached. Thus, besides the aforementioned time-dependence, stabilization of nonviral gene-delivery particles during freezing seems also to be highly temperature dependent. Similarly, Armstrong and Anchordoquy¹⁷ showed that particle aggregation does not occur during prolonged incubation in the frozen state at temperatures -22°C or below. Moreover, these results suggest that glass formation or vitrification of the sample is not required for gene-delivery-particle stabilization during freezing, which is in accordance with literature.⁷ However, the maintenance of the glassy state might be essential for the long-term storage stability of lyophilized formulations.^{2,45}

Furthermore, it was interestingly observed that partial slow freezing to a product temperature of $\sim -3^{\circ}\text{C}$ was as stressful as fast freezing to -45°C . This observation indicates that particle aggregation essentially occurs during the initial phase of ice formation and that continuing freezing at much lower temperature does not cause significant further damage as long as a sufficiently fast freezing rate is used.

This observed time dependence of particle destabilization during freezing indicates that destabilization of nonviral particles during freezing is to a marked degree kinetically controlled, and is likely diffusion controlled. As particle aggregation originates from colloidal interaction, particle mobility is a prerequisite for this bimolecular reaction. Particle mobility is strongly coupled with system viscosity and thus, particle stabilization can be achieved by a viscosity-driven decrease in the reaction rate constants.¹⁴ In general, according to Arrhenius kinetics, a decrease in temperature would decrease the bimolecular reaction rates.¹⁴ However, when ice formation occurs during freezing, solutes and particles present in the starting solution will experience a large increase in concentration. By the aid of our theoretical modeling, it can be illustrated that the sucrose concentration and consequently the system viscosity strongly increases, especially in the initial part of freezing during which the product temperature remains almost constant. This increase in the concentration of the solutes and particles in the freeze concentrate causes an increase in the rate of bimolecular reactions and thus, competes with the Arrhenius-dependent decrease in reaction rate, causing a maximum in reaction rate. To estimate the course of the bimolecular reaction rate of gene-delivery particles during freezing, a second-

order reaction with Arrhenius temperature dependence was assumed. Moreover, the reaction rate was coupled to viscosity assuming a diffusion-controlled reaction with the diffusion constant being inversely proportional to viscosity, as in the Stokes–Einstein equation.²⁴ It is certainly true that the calculated reaction rates are only rough estimations. But nevertheless, it is our belief that this model helps to illustrate the interrelation between bimolecular reaction rate and system viscosity, which is dependent on the freezing protocol and on initial solute concentration. During ice nucleation, a $\sim$ five-fold increase in the relative reaction rate was calculated for all samples. However, here, one has to keep in mind that all reaction rates are presented as relative reaction rates, which are related to the reaction rate at the point of ice nucleation at a DNA concentration of $50\text{ }\mu\text{g/mL}$. At this point, the liquid, ice-free sample has reached the lowest temperature during cooling before ice formation starts. Thus, the reaction rate that was used as reference is low compared, for example, with the reaction rate in the liquid state at room temperature. The reaction rate in a liquid sample at 20°C was determined from the model to be $\sim$ five fold higher compared with the reaction rate at -12°C . Thus, initial ice formation results in reaction rates that are comparable with reaction rates in liquid samples at room temperature. This indicates that it is not the initial formation of ice itself that is damaging but it is the freeze concentration.

When $50\text{ }\mu\text{g/mL}$ polyplexes were formulated at 10% sucrose, which is below the critical sucrose–DNA ratio, reaction rates increased as freezing progressed. However, in all cases, the maximum rate was reached early in the process at a rather high product temperature of $\sim -1^{\circ}\text{C}$ and a very low system viscosity of $\sim 7\text{ mPa s}$. Thus, when polyplexes are formulated at a too low initial sucrose concentration, bimolecular reactions will be accelerated during the very early part of freezing as the particle concentration increases, and the viscosity increase cannot compensate for this. However, as the product temperature further decreases, viscosity increases, substantially limiting particle diffusion and collision. At a certain point during freezing, this latter effect becomes dominant, resulting in a decrease in bimolecular reaction rates. In our model, the initial reaction rate was reached at a product temperature of $\sim -3^{\circ}\text{C}$ and viscosity of $\sim 50\text{ mPa s}$, indicating that the period before this point is the most critical temperature region during freezing. This prediction is in very good agreement with our experimental observations. Moreover, our model indicates that the bimolecular reaction rates approach zero at a temperature of -12°C and a viscosity of $\sim 16,500\text{ mPa s}$. This is also in good accordance to our experimental results as no further particle aggregation, even at very low shelf-ramp rates,

was found when the product temperature fell below between -11°C and -18°C . This highlights that in this temperature region, the system viscosity is already high enough to almost completely inhibit the particle mobility and that glass formation is not required, at least for times relevant to the freezing time scale. Armstrong and Anchordoquy¹⁷ reported that a viscosity of the freeze concentrate of 2.10×10^4 Pa s was required to prevent particle interaction when samples were incubated at -22°C for 24 h. But this study was performed at lower temperatures and a much lower sucrose–DNA ratio of 200 that was not capable of completely inhibiting particle aggregation during initial freezing. By the aid of our model, we could show that the point beyond which no further aggregation occurs is reached much earlier when freezing is performed at accelerated shelf-ramp rates. Furthermore, we could show that the initial stabilizer concentration strongly impacts the maximum reaction rates during freezing. When a sufficiently high initial excipient concentration is chosen, the increase in the bimolecular reaction rate during freezing after initial ice formation may be suppressed essentially completely. This result indicates that the initial system viscosity, which is directly related to the initial excipient concentration, is critical for the course of the reaction rate during freezing.

Allison et al.⁷ also mentioned that the high viscosity of the unfrozen fraction may serve to immobilize complexes and prevent aggregation. However, they reported that viscosity is not able to explain the observed critical sugar–DNA ratio, as the solute concentration in the freeze concentrate is only dependent on the sample temperature but not on the initial solute concentration. Thus, finally, they concluded that the distance between complexes in the unfrozen fraction, which is defined by the initial solute concentration, determines whether particles can physically interact or not.⁷ Besides the aforementioned results that do not fit to this theory, this hypothesis can be reinvestigated by the aid of our calculation procedure. For instance, the sucrose concentration in the unfrozen fraction of a 6% and 20% sucrose formulation is doubled to 12% and 40% at product temperatures of -0.4°C and -2.2°C , respectively. This small difference in product temperature leads to a substantial difference in the viscosity of this unfrozen fraction (3.2 vs. 29.4 mPa s) and in the reaction rate (~ 16 vs. ~ 3). Thus, although the same volume of the unfrozen fraction is given, particle stability differs considerably mainly because of the clear difference in sample viscosity between small differences in sample temperature.

In contrast to our observations on the stability of polyplexes, a better stability of lactate dehydrogenase was reported when samples were frozen with the “standard” depressurization method in comparison with shelf-ramp freezing.¹⁹ However, Bhatnagar

et al.²³ showed that ice formation and not freeze concentration is the critical destabilizing factor during freezing of lactate dehydrogenase. Thus, this comparison shows that an optimized freezing procedure cannot simply be directly transferred to another pharmaceutical active ingredient. Rather, an extended understanding of the involved destabilization mechanism is necessary to design an optimal freezing process during lyophilization, which will vary from case to case. However, we believe that the qualitative results of this study should also apply to other systems that are prone to aggregation such as protein formulations. But when applying this approach to other systems, the quantitative outcome will be affected by the difference in particle size.

SUMMARY

The aim of the study was to provide an extended understanding of critical freezing parameters and underlying stabilization mechanisms of nonviral gene-delivery particles during the freezing step of lyophilization. In general, it was observed that particle size of plasmid/LPEI polyplexes is better preserved with an increase in sucrose–DNA ratio but the critical ratio above which no particle aggregation occurs is highly dependent on the freezing process used. To explain this observation, the influence of ice-nucleation temperature and residence time was investigated separately by the aid of the easy-to-apply depressurization method. Only the stability of particles at low concentration correlated negatively with an increase in ice-nucleation temperature. For both DNA concentrations, a more distinct increase in particle size was detected when samples were frozen at decreased shelf-ramp rates after controlled ice nucleation. Depressurization in combination with fast subsequent ramp rates resulted in comparable particle sizes as conventional shelf-ramp freezing, indicating that a temperature-hold step after ice nucleation is damaging with regard to particle stability. Destabilization of particles during freezing occurs predominantly during the initial temperature decrease down to $\sim -3^{\circ}\text{C}$ after ice nucleation. In this phase, cryoconcentration has already proceeded greatly, but sample viscosity is still too low to fully inhibit particle mobility. This experimental observation matched well with calculations using our theoretical model. Here, a maximum of the reaction rates was found at a product temperature of $\sim -1^{\circ}\text{C}$, and reaction rates dropped to reach their initial levels at product temperatures of $\sim -3^{\circ}\text{C}$. No further particle aggregation was observed at temperatures below $\sim -18^{\circ}\text{C}$, indicating that at this temperature and degree of freeze concentration, particle mobility is sufficiently inhibited with reaction rates close to zero, despite the absence of glassy state, due to the high sample viscosity.

Furthermore, our model calculations showed that the level of the maximum reaction rate is determined by the initial sucrose–DNA ratio. This fully explains the need of a critical sucrose–DNA for complete particle stabilization.

CONCLUSIONS

Our results show that the time that a particle spends in the low-to-moderate viscous state after ice nucleation is the predominant factor in the stabilization of nonviral gene-delivery particles during freezing. This residence time is influenced by the initial sample viscosity, which is directly related to the initial excipient concentration, but also controlled by the applied freezing rate. This extended understanding of the relevant stabilization mechanisms during freezing will facilitate the development of optimized formulations and less-stressful lyophilization processes, always to be performed hand-in-hand. Finally, although the quantitative results obtained are specific to the system studied, most of the qualitative results should apply to other systems that degrade by aggregations, including proteins.

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